

Discovering Latent Musical Structure Using Spotify Audio Features

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I. INTRODUCTION

This project explores latent musical structure using Spotify audio features through data mining techniques. The focus is on discovering interpretable patterns rather than building predictive models. This milestone (M2) emphasizes exploratory data analysis, preprocessing, and initial clustering.

II. EXPLORATORY DATA ANALYSIS (EDA)

A. Dataset Overview

The dataset contains approximately 114,000 Spotify tracks with 21 features, including both metadata and numerical audio attributes such as danceability, energy, loudness, acousticness, instrumentalness, valence, tempo, duration, and popularity.

B. Data Quality Assessment

- Total observations: 114,000 tracks
- Total features: 21 variables
- Duplicate rows: 0
- Missing values: 3 (metadata only)

The missing values occur in metadata fields (artist, album name, track name). Importantly, no missing values are present in the numerical features used for clustering, so no imputation was required. This confirms that the dataset is clean and suitable for analysis.

C. Distributional Patterns

Exploratory visualizations (histograms, boxplots, and correlation heatmaps) revealed several key patterns:

- Energy and loudness show strong positive correlation, reflecting production intensity.
- Acousticness is negatively correlated with energy and loudness, indicating a contrast between acoustic and highly produced tracks.
- Popularity exhibits weak correlation with most audio features, suggesting that commercial success is not strongly driven by any single musical dimension.
- Features such as danceability, energy, and valence are already scaled between 0 and 1, while duration and tempo have larger numeric ranges.

Overall, feature distributions suggest gradual variation rather than sharply separated categories, indicating that musical styles exist along continuous dimensions.

III. DATA PREPROCESSING

A. Feature Selection

Only numerical features were retained for clustering. The following columns were removed:

- Unnamed: 0 (index column)
- Key, mode, time_{signature}(*categoricalvariables*)

These categorical variables were excluded to avoid imposing artificial ordinal structure.

The final feature set includes:

- Popularity
- Duration (ms)
- Danceability
- Energy
- Loudness
- Speechiness
- Acousticness
- Instrumentalness
- Liveness
- Valence
- Tempo

B. Feature Scaling

Standardization (Z-score normalization) was applied using `StandardScaler`:

- Mean ≈ 0
- Standard deviation ≈ 1

Scaling ensures that features with larger ranges (e.g., duration, tempo) do not dominate distance calculations in clustering.

IV. FIRST MINING RESULTS

A. Clustering Method

K-Means clustering was implemented as the primary mining technique.

B. Selecting Number of Clusters

Two evaluation methods were used:

1) *Elbow Method*: The inertia curve shows a sharp decrease from $k = 1$ to $k = 3$, followed by diminishing returns. This suggests that 3–4 clusters are appropriate.

2) Silhouette Score:

- $k = 2 \rightarrow 0.1889$
- $k = 3 \rightarrow 0.0931$
- $k = 4 \rightarrow 0.0933$
- $k = 5 \rightarrow 0.0992$
- $k = 6 \rightarrow 0.0919$

Although $k = 2$ yields the highest score, it produces overly coarse segmentation. Therefore, $k = 3$ was selected as a balance between interpretability and structure.

C. Cluster Characterization

Three interpretable clusters were identified:

1) Cluster 0: Energetic / Dance-Oriented Tracks:

- Energy ≈ 0.69
- Danceability ≈ 0.68
- Acousticness ≈ 0.27
- Tempo ≈ 118 BPM
- Popularity ≈ 32.9

These tracks are rhythm-driven and production-heavy, likely corresponding to energetic popular music.

2) Cluster 1: Acoustic / Low-Energy Tracks:

- Acousticness ≈ 0.77
- Energy ≈ 0.28
- Loudness ≈ -14.77 dB
- Tempo ≈ 109 BPM
- Instrumentalness ≈ 0.32

These songs are softer, more acoustic, and less production-intensive.

3) Cluster 2: High-Intensity / Fast-Paced Tracks:

- Energy ≈ 0.80
- Tempo ≈ 135 BPM
- Acousticness ≈ 0.09
- Danceability ≈ 0.48

These tracks represent high-tempo, high-intensity music, potentially electronic or festival-oriented.

D. Overall Interpretation

Clustering primarily separates songs based on:

- Energy
- Acousticness
- Tempo
- Production intensity

Clusters are not defined by popularity, reinforcing that commercial success is not a dominant structural factor.

PCA visualization shows:

- Visible but overlapping clusters
- Gradual transitions between styles
- Continuous rather than discrete structure

This aligns with the relatively low silhouette scores, indicating that musical features vary along smooth latent dimensions.

V. PLAN FOR M3

Based on M2 findings, the following steps are planned:

- Apply DBSCAN to identify non-spherical clusters
- Compare density-based clustering with K-Means
- Perform anomaly detection to identify unusual tracks
- Evaluate clustering robustness using additional metrics
- Explore alignment with genre labels

VI. CONCLUSION

K-Means clustering identifies three meaningful musical archetypes based on intensity, acoustic characteristics, and tempo using 114,000 Spotify tracks and 11 numerical features.

The results demonstrate that Spotify audio features encode interpretable latent musical structure, although cluster separation is moderate. These findings provide a strong foundation for further analysis using density-based clustering and anomaly detection in M3.